Frank Poon

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Experience

META

Software Engineer • *October 2024 – Present* • *New York, NY, USA* • *meta.com*
Currently under Meta Capacity Planning (MCP), building tools to help Meta forecast future infrastructure needs in line with predicted product growth.

- Design and implement revamp of validation service for forecasting data, lowering average validation run latency by 80% and facilitating an organization-wide rollout of the service to 3 teams/domains within MCP
- Enable decimal precision across entire forecasting system, allowing for cost savings of $117 million across all of Meta on storage expenditures from using more precise figures

TIKTOK (BYTEDANCE)

Software Engineer • *July 2022 – July 2024* • *New York, NY, USA* • *tiktok.com*
Worked under TikTok's Creator organization, improving content quality and audience engagement.

- Led development and deployment of ML model infrastructure for plant scanner sticker feature, surpassing expected adoption rate by 60%
- Improved data ecosystem for paid content conversion funnel by identifying and fixing misreported metrics to ensure accurate business insights
- Executed improvements to paid content product based on data analysis, increasing return on ad spend from 60% to 80%
- Mentored junior engineers and organized knowledge sharing sessions to facilitate organizational growth from 20 to 40 members

AMAZON

Software Development Engineer • *August 2020 – June 2022* • *Cambridge, MA, USA* • *alexa.amazon.com*
Alexa Secure AI Foundations (SAIF) makes Alexa data more accessible to internal modelers, data scientists, and engineers in both quality and quantity.

- Spearheaded organization-wide privacy mitigation effort, unblocking over 1000 DS/SDEs in 16 Alexa AI teams by enabling use and export of pseudonymized datasets with a differential privacy algorithm
- Reduced average and query time for privacy-critical Alexa data from 90 min to 30 sec
- Increased CI/CD pipeline deployment availability and success rates from 40% to 90%

Volunteer Work

COMMUNITY BOATING INC.

Volunteer Software Engineer • *January 2022 – April 2022* • *Boston, MA, USA* • *community-boating.org*
Community Boating is the oldest public sailing organization in the United States, with a mission to allow people of all backgrounds and abilities to sail.

- Developed notification feature with customizable alerts on sailing conditions
- Conducted traffic estimation and wrote technical design with API contract and business logic
- Implemented frontend; reviewed backend code and DB schema; performed integration testing

Skills

Languages: Python, Go (Golang), Java, JavaScript, Typescript, C/C++, PHP, bash, HTML, CSS, SQL
Technologies/Concepts: REST, RPC, node.js, React, Agile/Scrum, Distributed Systems
Tools: git, AWS, Google Cloud (GCP), SQL, Redis, Docker, Kubernetes

Education

UNIVERSITY OF WASHINGTON

B.S. Computer Science • *September 2016 – June 2020* • *Seattle, WA, USA*